

5 Dataset and Experimental Setup

5.1 Datasets

Training problems. For RL we use `train_30k`, the 30,000-problem file that Hou et al. (2025) release with their training script. Each problem comes with a final answer and no reference solution. By source tag, it contains WebInstruct (26%), NuminaMath (24%) and its `cn_k12` subset (17%), Omni-MATH (13%), and smaller sources.

A tree whose leaves are all correct or all wrong yields zero advantage under tree-based credit, so it contributes no gradient. We therefore filter by difficulty. We sample 3,000 problems and let the base model answer each 8 times (temperature 1.0): 42% are never solved and 10% always.

We keep the 1,432 problems solved 1–7 times out of 8. Our long RL runs use 1,200 of them (150 steps $\times$ 8 problems), each seen once.

Inference-only study (Phase I). The EPTree study in Section 6.1 needs no training data. It uses 500 Omni-MATH problems (Gao et al., 2024), a random sample (seed 0) without any filtering, which is the same construction as the paper’s Omni-MATH-500. We chose hard, unfiltered problems because PassRate (at least one correct response) saturates on easier sets: an earlier run on 243 accuracy-filtered OlympiadBench problems reached 95% PassRate with 16 i.i.d. chains, leaving no room to compare samplers. On our Omni-MATH sample the base model solves 26% of problems with one response and 53% with 64.

Evaluation benchmarks. We evaluate on three sets, 1,082 problems in total:

- MATH500 (Hendrycks et al., 2021; Lightman et al., 2024): 500 problems.
- AMC: 83 problems from the AMC 12A/12B contests of 2022 and 2023.
- Omni-MATH-500: a random sample (seed 0) of 500 Omni-MATH problems (Gao et al., 2024). We score 499 of them; one prompt exceeds the 1,024-token limit.

Omni-MATH-500 overlaps `train_30k`: 14 of its problems, and 1 MATH500 problem, are among our 1,200 training problems. We report results with these 15 problems removed as well.

Reward and grading. The reward is binary: 1 if the last `\boxed{}` answer matches the reference under `math-verify`, else 0. A response cut off at the length limit counts as wrong.

5.2 Model and Training

Model. All experiments use Qwen2.5-Math-1.5B-Instruct (Yang et al., 2024) with its default chat template. We sample at temperature 1.0 and top- p 0.95. At the paper’s temperature of 1.2, this model produced degenerate output in 13% of samples. Responses are capped at 3,072 tokens for the Phase I study and 2,048 for RL; only 1.1% of the model’s responses exceed 2,048 tokens.

Training stack. We reuse the TreeRL authors’ released implementation unchanged for the learning algorithm itself: EPTree tree construction, tree-derived advantages and the policy-gradient loss. We port its infrastructure to a single 46 GB GPU:

- the policy (DeepSpeed ZeRO-2, CPU Adam) and the vLLM sampler share the GPU;
- updated weights reach vLLM through CUDA IPC instead of NCCL;
- the trees of all problems in a step are sampled in one batched vLLM call.

Peak GPU memory is 18.5–21 GB, so 1.5B-parameter RL fits comfortably in a 40 GB budget.

Methods compared. All three methods share the same code path and differ only in how responses are sampled and credited:

- **TreeRL:** EPTree with $(M, N, L, T) = (6, 2, 1, 2)$, i.e. 30 leaves per problem. 8 leaves are trained on, with TreeRL’s tree advantages (global plus local value differences, scaled by $1/\sqrt{|\mathcal{L}(v)|}$).
- **ChainRL:** 8 independent chains through the same code. Without branching, the tree advantage reduces to a leave-one-out (RLOO) baseline.
- **GRPO** (Shao et al., 2024): the same 8 chains with the group-normalised advantage $(r_i - \bar{r})/\text{std}(r)$. The KL penalty is set to $\beta = 0$, as for the other two methods.

Table 1 lists the settings next to the paper’s.

Evaluation protocol. We report greedy accuracy and `mean@8`, the accuracy averaged over 8 samples per problem at temperature 1.0. Because scoring the same model twice varies by about 0.5 points, models are compared on the same problems with a paired bootstrap: 2,000 resamples of problems, 95% CI.

6 Results

6.1 Phase I: surprisal-guided exploration

Phase I of our method builds the exploratory tree with EPTree’s surprisal criterion. We first replicate EPTree

	TreeRL paper	Ours
Policy	7–14B	Qwen2.5-Math-1.5B
RL steps	~300	40 / 150
Learning rate	1.5e-6	1.5e-6 / 5e-6
Problems / step	16	8
Trained resp./problem	30 Tree, 16 Chain	8
KL β	10^{-4}	0
Temperature, top- p	1.2, 0.95	1.0, 0.95
Max response tokens	8,192	2,048
Optimiser	ZeRO-3, CPU Adam	ZeRO-2, CPU Adam

Table 1: RL settings: the TreeRL paper and ours. Each of our steps uses 64 trained responses and one optimiser update.

as an inference-only study on 500 Omni-MATH problems, using TreeRL’s own tree construction code. We compare three samplers:

- EPTree, which forks at the N highest-surprisal tokens;
- random forking with the same (M, N, L, T) , the paper’s ablation;
- i.i.d. chains, scored with the unbiased $\text{pass}@k$ estimator from 64 samples per problem.

Trees beat i.i.d. sampling at the same cost. Every tree configuration lies above the i.i.d. curve (Figure 1a). The gain at a matched token budget is +0.6 to +3.3 points, and significant for $(2, 3, 1, 1)$ and $(6, 2, 1, 2)$ (Figure 1b, Table 2). This replicates the paper’s main inference result.

Where to fork matters little for PassRate. At identical (M, N, L, T) , entropy-guided and random forking differ by +0.4, +0.8, 0.0 and -0.8 points, and every CI includes zero. The paper’s differences (+2.1, +1.0) lie inside our intervals. In this setting the gain over i.i.d. sampling comes mostly from sharing prefixes, not from choosing the fork point.

Surprisal finds uncertain tokens, rarely consequential ones. EPTree does fork where the model is uncertain: the forked token has a median surprisal of 3.56 nats, while 63% of random forks land on near-certain tokens (Figure 2a). Its branches also end in a different final answer more often than random branches: 46.6% vs. 42.6%, paired difference +4.0 [+3.2, +4.8].

However, a branch changes correctness, the evidence that credit assignment needs, in only 7.0% of cases for EPTree and 6.1% for random forks (+0.9 [+0.5, +1.3]). Over 90% of rollouts reproduce their parent’s outcome. Fork positions are spread across the response (mean relative position 0.50), and the most frequent fork tokens are function words and $\LaTeX$ delimiters.

This is the gap our method targets. Surprisal is a reasonable exploration signal, but most of a fixed budget

spent on it yields no counterfactual evidence. Phase II therefore spends the reserved k rollouts only on states that are both novel and consequential.

6.2 RL baselines

Short runs. With 40 steps at the paper’s learning rate, none of the three methods changes the model measurably. On greedy MATH500, for example, step-40 checkpoints score 74.4–75.4 against the base model’s 74.6, and every difference across the three benchmarks is within noise. We therefore ran 150 steps with a higher learning rate (5×10^{-6}).

Long runs. TreeRL and ChainRL are indistinguishable at steps 50 and 100 (Figure 3a). By step 150, TreeRL improves over the base model by +0.92 points [+0.17, +1.66], while ChainRL stays at the base level (-0.16 [-0.85 , +0.52]).

Head-to-head on the same problems, TreeRL leads ChainRL by +1.09 [+0.32, +1.84] (Figure 3b, Table 3). It is ahead on every benchmark: MATH500 +1.07, AMC +2.41, Omni-MATH-500 +0.88. Excluding the 15 test problems that overlap the training data gives +1.04 [+0.27, +1.79].

The pattern matches the paper, where both methods track each other early and TreeRL pulls ahead after about 100 of ~300 steps. The size of the gap is close to the paper’s for an already strong policy: +1.3 for R1-Distill-Qwen-7B.

Caveats.

- Each method was trained once, so run-to-run variance is not yet measured.
- The methods are matched on trained responses, not on generated tokens. Our TreeRL generates about $2.4\times$ more tokens than ChainRL, whereas the paper matches generated tokens.
- Our budget (150 steps $\times$ 64 responses) is roughly 1/15 of the paper’s. The base policy, already tuned for math by its developers, leaves little headroom on MATH-level problems.
- The GRPO baseline has finished 50 of its 150 steps.

References

Bofei Gao, Feifan Song, Zhe Yang, Zefan Cai, Yibo Miao, Qingxiu Dong, Lei Li, Chenghao Ma, Liang Chen, Runxin Xu, et al. Omni-MATH: A universal olympiad level mathematic benchmark for large language models. *arXiv preprint arXiv:2410.07985*, 2024.

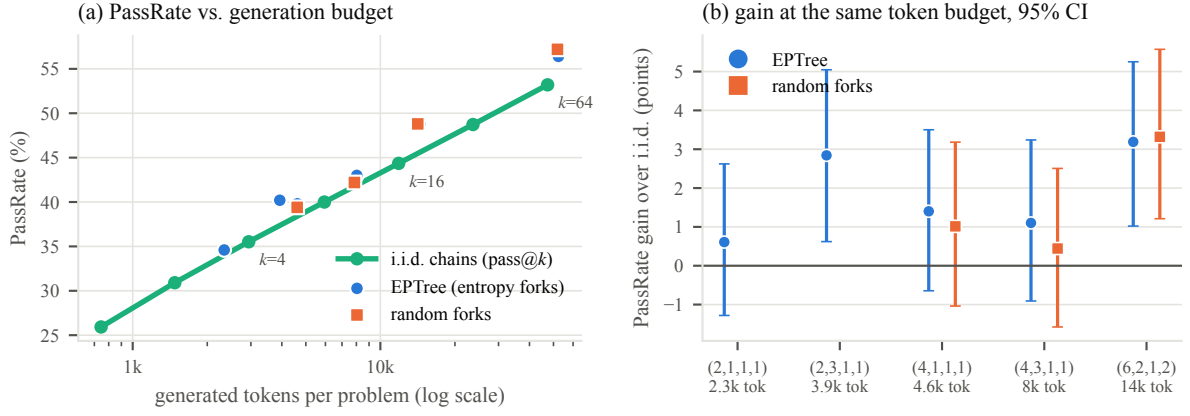


Figure 1: Tree sampling vs. i.i.d. chains on 500 Omni-MATH problems. (a) PassRate (at least one correct response) against generated tokens. (b) Gain over i.i.d. chains at the same token budget, per problem with a 95% bootstrap CI. The i.i.d. PassRate is interpolated log-linearly in tokens between neighbouring k . (8, 4, 2, 2) is omitted because it exceeds the 64-chain budget.

(M, N, L, T)	Leaves	Tokens	EPTree	Random	i.i.d. at same tokens	EPTree — i.i.d. [95% CI]	EPTree — Random [95% CI]
(2,1,1,1)	4	2.3k	34.6	—	34.0	+0.6 [−1.3, +2.6]	—
(2,3,1,1)	8	3.9k	40.2	—	37.4	+2.8 [+0.6, +5.0]	—
(4,1,1,1)	8	4.6k	39.8	39.4	38.4	+1.4 [−0.6, +3.5]	+0.4 [−2.0, +2.8]
(4,3,1,1)	16	8.0k	43.0	42.2	41.9	+1.1 [−0.9, +3.2]	+0.8 [−1.6, +3.4]
(6,2,1,2)	30	14.5k	48.8	48.8	45.6	+3.2 [+1.0, +5.3]	0.0 [−2.4, +2.4]
(8,4,2,2)	136	52.4k	56.4	57.2	—	—	−0.8 [−3.2, +1.6]

Table 2: PassRate (%) of tree sampling on 500 Omni-MATH problems. Tokens are generated tokens per problem, counted once per shared prefix. The i.i.d. column is pass@ k at the same token budget. i.i.d. chains reach 44.4% at 16 chains and 53.2% at 64.

	MATH500	AMC	Omni	Pooled	Mean
Base	73.95	45.03	26.68	49.93	48.55
ChainRL	73.83	45.78	26.33	49.77	48.65
GRPO	in progress				
TreeRL	74.90	48.19	27.20	50.85	50.10
TreeRL — ChainRL, pooled: +1.09 [+0.32, +1.84]					

Table 3: Mean@8 accuracy (%) at step 150 of the long runs. Pooled: over all 1,082 problems. Mean: unweighted mean of the three benchmarks, as the paper reports.

Dan Hendrycks, Collin Burns, Saurav Kadavath, Akul Arora, Steven Basart, Eric Tang, Dawn Song, and Jacob Steinhardt. Measuring mathematical problem solving with the MATH dataset. In *Advances in Neural Information Processing Systems (NeurIPS) Datasets and Benchmarks Track*, 2021.

Zhenyu Hou, Ziniu Hu, Yujiang Li, Rui Lu, Jie Tang, and Yuxiao Dong. TreeRL: LLM reinforcement learning with on-policy tree search. In *Proceedings of the 63rd Annual Meeting of the Association for Computational Linguistics (ACL 2025)*, 2025.

Hunter Lightman, Vineet Kosaraju, Yura Burda, Harri

Edwards, Bowen Baker, Teddy Lee, Jan Leike, John Schulman, Ilya Sutskever, and Karl Cobbe. Let’s verify step by step. In *International Conference on Learning Representations (ICLR)*, 2024.

Zhihong Shao, Peiyi Wang, Qihao Zhu, Runxin Xu, Junxiao Song, Xiao Bi, Haowei Zhang, Mingchuan Zhang, Y. K. Li, Y. Wu, and Daya Guo. DeepSeek-Math: Pushing the limits of mathematical reasoning in open language models. *arXiv preprint arXiv:2402.03300*, 2024.

An Yang, Beichen Zhang, Binyuan Hui, Bofei Gao, Bowen Yu, Chengpeng Li, Dayiheng Liu, Jianhong Tu, Jingren Zhou, Junyang Lin, et al. Qwen2.5-Math technical report: Toward mathematical expert model via self-improvement. *arXiv preprint arXiv:2409.12122*, 2024.

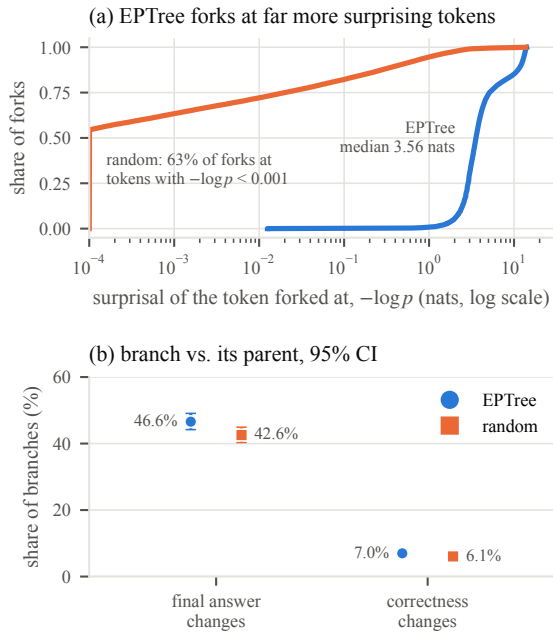


Figure 2: Fork points of EPTree vs. random forking (four tree shapes run with both, 84k forks each). (a) Distribution of the surprisal of the token forked at. (b) Share of branches whose final answer, or whose correctness, differs from their parent’s, per problem with a 95% bootstrap CI.

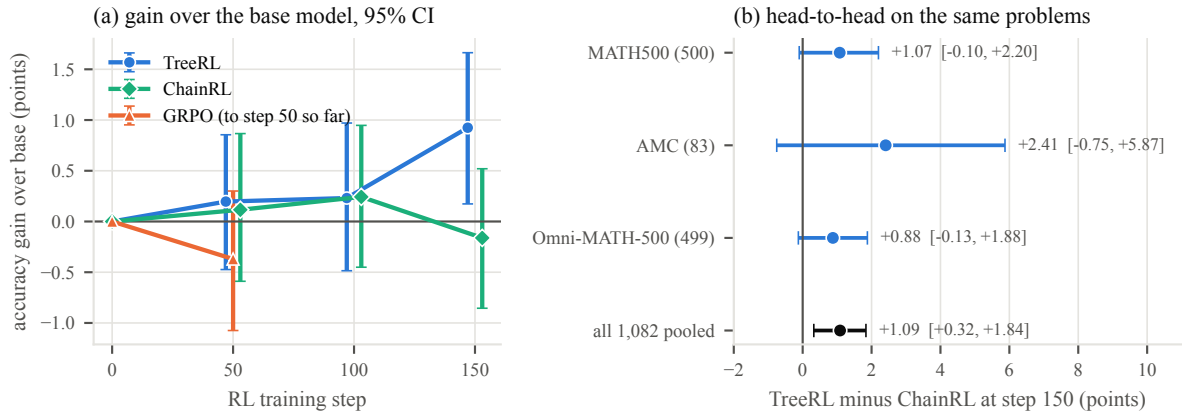


Figure 3: RL with Qwen2.5-Math-1.5B-Instruct: 150 steps, 8 problems $\times$ 8 trained responses per step. Accuracy is mean@8 on the same 1,082 test problems, with paired bootstrap 95% CIs. (a) Gain over the base model during training. GRPO is evaluated up to step 50 so far. (b) TreeRL minus ChainRL at step 150, per benchmark and pooled.